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Heart rate variability, eye temperature, and spontaneous blink rates as measures of emotions in working donkeys

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Abstract

Quantitative measures of emotion play a crucial role in objectively assessing animal welfare. However, there is currently no research available on donkey emotions. Our study investigated spontaneous blink rate (SBR), eye temperature (ET), and heart rate variability (HRV), quantified as RMSSD (root mean square of the successive differences between beat-to-beat intervals), as potential measures of donkey emotions. Thirty-four donkeys were exposed to three treatments: positive, negative, and control. The positive and negative treatments were feeding high-quality lucerne and exposing donkeys to the sound of hair clippers, respectively. The control group was recorded standing in their usual environment, and were not subjected to any additional positive or negative stimuli. Data were analyzed using linear models. SBR increased during positive treatment and decreased with the negative treatment compared to the control. An increase in ET was found for both negative and positive treatment compared to the control. We found a positive relationship between changes in SBR and HRV (estimated=1.32, t=5.44, p<0.001), whereas negative relationships were found between changes in ET and HRV (estimated= -12.76, t= -5.93, p<0.001). ET, SBR, and HRV may act as indicators of emotions in donkeys. Additional studies should investigate these metrics through a thorough analysis of donkey behavior.

Keywords: animal sentience, cardiology, physiology, thermography, welfare, working equids

1. Introduction

Animal emotion is a complex response comprising subjective, physiological, neural, and cognitive components to a stimulus or event [1,2]. Emotions play a crucial role as they facilitate the selection of appropriate behavioral responses to external and internal events [3]. Emotions have two main dimensions: arousal (calm versus excited) and valence (negative or positive) [4,5]. The existence and nature of emotions in non-human animals are believed to be correlated with those in humans and to have evolved from similar mechanisms [6]. However, emotions are subjective internal experiences that cannot be easily measured [7]. Consequently, techniques for understanding and assessing animal emotions, particularly positive ones, are limited [3]. Currently, the focus lies in inferring emotions by analyzing behavioral,

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physiological, and neurobiological indicators [7,8]. Animal emotions change in response to positive and negative stimuli [9,10]. Therefore, accurately inferring animal emotional states from behavioral and physiological evidence is important for welfare, because emotions reflect how animals experience their lives and influence how they respond to environmental challenges and resources [11,12]. There has been an increasing interest in identifying non-invasive and reliable indicators of both positive and negative emotions in animals [9,10,13].

Globally, there are around 50.5 million donkeys (*Equus asinus*) [14], and they support approximately 600 million people, especially in low and middle-income countries (LMICs) where they are relied upon for agriculture, construction, and transportation of goods [15,16]. Despite their importance, relatively little research has been conducted to assess and improve the welfare of working donkeys in LMICs [17]. This has led to their neglect in government-level animal welfare regulations [18], although working equids were recently included in the World Organization for Animal Health's (WOAH) Terrestrial Animal Welfare Code [19]. Welfare challenges for working donkeys in LMICs are multifactorial, driven by interacting socioeconomic and structural factors (poverty, limited access to education and veterinary care, reliance on donkeys for livelihoods, and inadequate legal or welfare protections). Together, these factors can lead to harsh working conditions and harm donkey health and welfare [12]. This is particularly evident in donkeys used for carrying heavy loads when owners may overlook their emotions and pain [20].

Spontaneous blink rate (SBR) has been utilized to measure stress responses in humans [21] and offers a non-invasive method for assessing stress in animals [13,22,23]. Blinks serve multiple functions, including corneal lubrication and the resetting of eye movements during fixation error correction [24]. The association between SBR and stress appears to be influenced by dopaminergic activity. In monkeys, SBR increased following subcutaneous administration of dopamine agonists and was reduced with dopamine antagonists [25]. Humans demonstrate a decrease in SBR during periods of focused attention and cognitive effort [21]. SBR is a reliable measure of stress in humans and animals [10,13,21]. The use of SBR to assess stress or negative emotional state is promising, as it is a non-invasive and inexpensive measure of stress [10]. Once investigated and standardized under donkey working conditions, it could help donkey handlers, owners, and veterinarians to make better decisions for their care and ultimately improve the welfare of animals.

Eye temperature (ET) measurement effectively evaluates animal stress across a range of species [26–29]. This method involves using infrared thermography to assess the temperature of the eye, usually at the medial canthus, which can reflect the physiological and emotional state of horses (*Equus caballus*) [30], goats (*Capra hircus*) [31], sheep (*Ovis aries*) [32], cats (*Felis catus*) [33], dogs (*Canis familiaris*) [34], and cattle (*Bos taurus*) [35]. Changes in eye temperature can be influenced by the proximity of the orbit to the brain and changes in blood flow caused by vascular activity, which are commonly affected by stress [36]. A positive relationship between eye temperature (ET) and salivary cortisol levels has been reported [29]. Moreover, a negative relationship between ET and heart rate variability (HRV) was observed in response to stress [26]. The use of ET as a non-invasive measure of stress has great potential for practical application. However, there is currently no research available on how the ET of donkeys varies during positive and negative emotional states.

HRV is the physiological phenomenon of variations in the time interval between heartbeats [3,26] and can be measured as root mean square of the successive differences of the beat-to-beat interval, recorded in milliseconds (RMSSD) [10]. It represents the ability of the heart to respond to a variety of physiological and environmental stimuli [37]. RMSSD is a valuable tool for assessing animal stress and positive emotional states [3,10]. The interplay between the sympathetic and parasympathetic branches of the nervous system influences the HRV [38,39] and controls the physiological responses of animals to positive and negative stimuli

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[40–43]. Animals in a relaxed and comfortable state exhibit increased RMSSD [40,44], whereas those experiencing stressful situations exhibit decreased HRV [26,45,46]. This decrease in RMSSD indicates an increase in sympathetic activity and a decrease in parasympathetic influence, reflecting a state of heightened stress [10,26,45,46]. Moreover, it is reported that animals engaging in positive interactions exhibit increased RMSSD [40,44]. This suggests a higher parasympathetic influence and a state of relaxation and contentment [40]. However, RMSSD should be interpreted cautiously because it is influenced by the specific metric used, heart rate, respiration, physical activity, and the level of physiological arousal. While RMSSD is a promising measure of stress and positive emotions in animals (e.g., goats [3,38], horses [10,47]), donkeys have not yet been investigated in this regard.

Donkeys are known for their stoic behavior [48,49] and typically exhibit a reduced flight response and tend to show freeze behaviors more often than flight behaviors when compared to horses [48,49]. Many people believe that donkeys feel less pain and stress because they hide signs of pain, making it difficult to assess their welfare [50]. However, this perception is due to a lack of understanding of donkeys and their emotional states, which can greatly affect their welfare. Recent research has shown that recognizing donkey emotions and feelings of pain (i.e., recognizing that donkeys are sentient) is associated with improved management practices [20]. Therefore, it is necessary to investigate the indicators of positive and negative emotions in donkeys to understand and protect their welfare.

Our research investigated how SBR, ET, and HRV change in response to positive and negative treatment exposure in donkeys. We hypothesized that positive treatment would lead to a greater change in SBR than negative treatment. In contrast, negative treatment would result in a greater change in ET than positive treatment, and the change in HRV would be greater during positive treatment than during negative treatment.

2 Results

2.1 Level of agreement and bias of the spontaneous blink rate between coders

The agreement for SBR between two coders was high, with a mean bias of − 0.09 to − 0.17 and narrow limits of agreement. Bias during control, positive, and negative treatment was -0.09, -0.15, and -0.17, respectively. The limits of agreement were narrow (less than the SBR standard deviations) in all situations, Figure 1.

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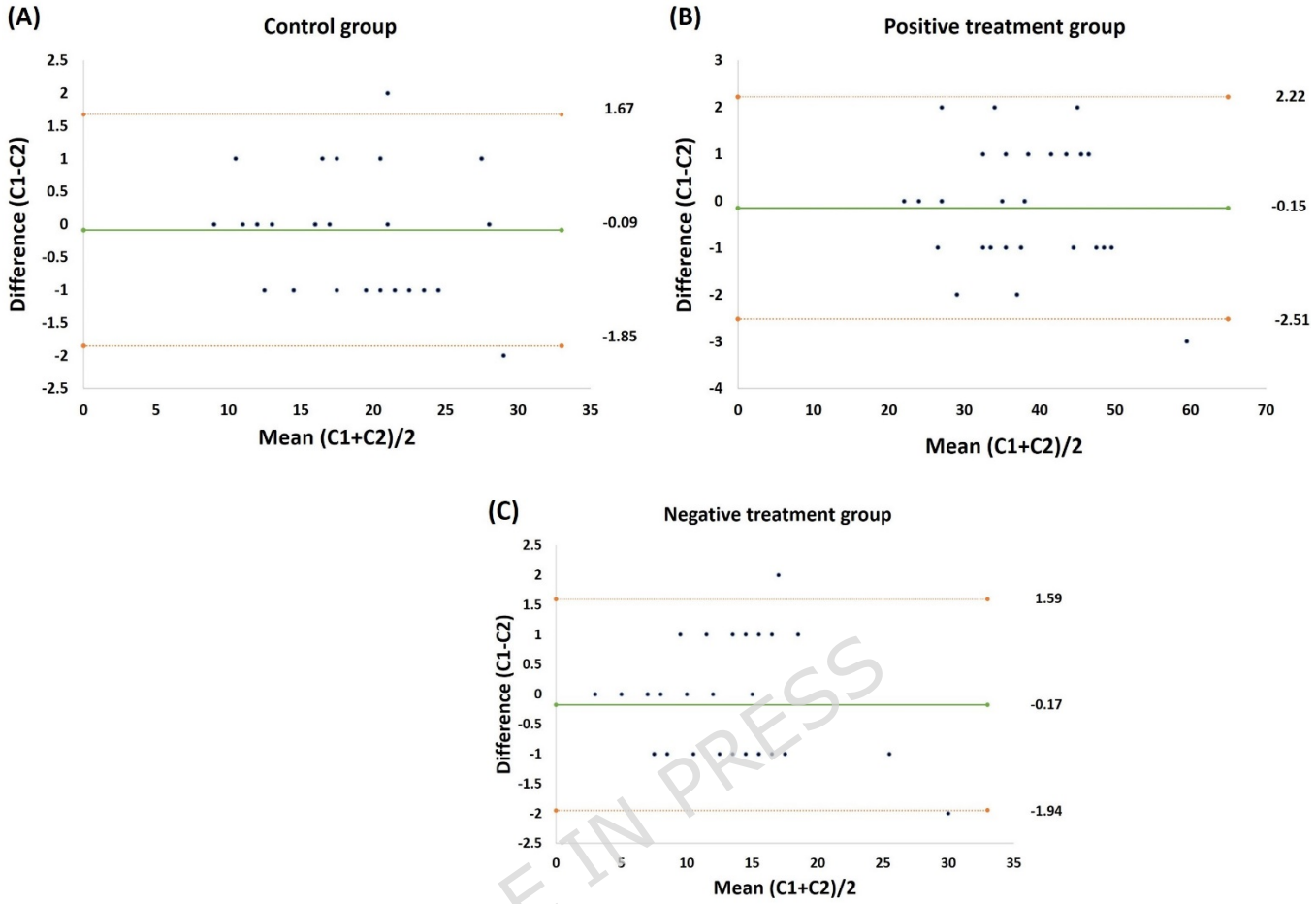


Figure 1: Bland and Altman plots showing the agreement and bias of the spontaneous blink rate (blinks per min) of donkeys between coders (C1 and C2) during the (A) control (B) positive, and (C) negative treatment. The bias (green continuous line) and limits of agreement (orange dotted lines) are indicated on each plot.

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2.2 Donkey body condition score, ambient temperature, humidity, and wind speed

The mean ambient temperature was 25.65°C, and the mean humidity was 34.64%. The mean wind speed (ms^{-1}) on the working days was 0.21, Table 1.

Table 1: Donkey body condition score, ambient temperature, humidity, and wind speed are presented as mean and standard deviation (mean ± SD).

Variables	Mean ± SD
Ambient temperature (°C)	25.65 ± 4.55
Humidity (%)	34.64 ± 5.79
Wind speed (km/h)	0.78± 0.31

2.3 Change in spontaneous blink rate, eye temperature, and heart rate variability during the positive and negative treatments

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Descriptive changes in SBR, ET, and HRV (RMSSD) during positive and negative treatments are presented in Table 2. The mean change in SBR was 19.0 ± 6.62 blinks per min during the positive treatment and 6.79 ± 3.17 blinks per min during the negative treatment. The mean change in ET was 0.81 ± 0.56 °C during the positive treatment and 1.22 ± 0.57 °C during the negative treatment. The mean change in HRV (RMSSD) was 37.73 ± 15.95 ms during the positive treatment and 10.73 ± 7.04 ms during the negative treatment.

Table 2: Spontaneous blink rate, eye temperature, and heart rate variability (RMSSD) during control, positive, and negative treatment of donkeys. Table 2 shows the mean and standard deviation (mean ± SD) of the absolute values and changes (control minus positive and control minus negative treatment) in each variable during positive and negative treatment. Heart rate variability was quantified using the root mean square of the successive differences between beat-to-beat intervals (RMSSD), a time-domain HRV metric reported in milliseconds.

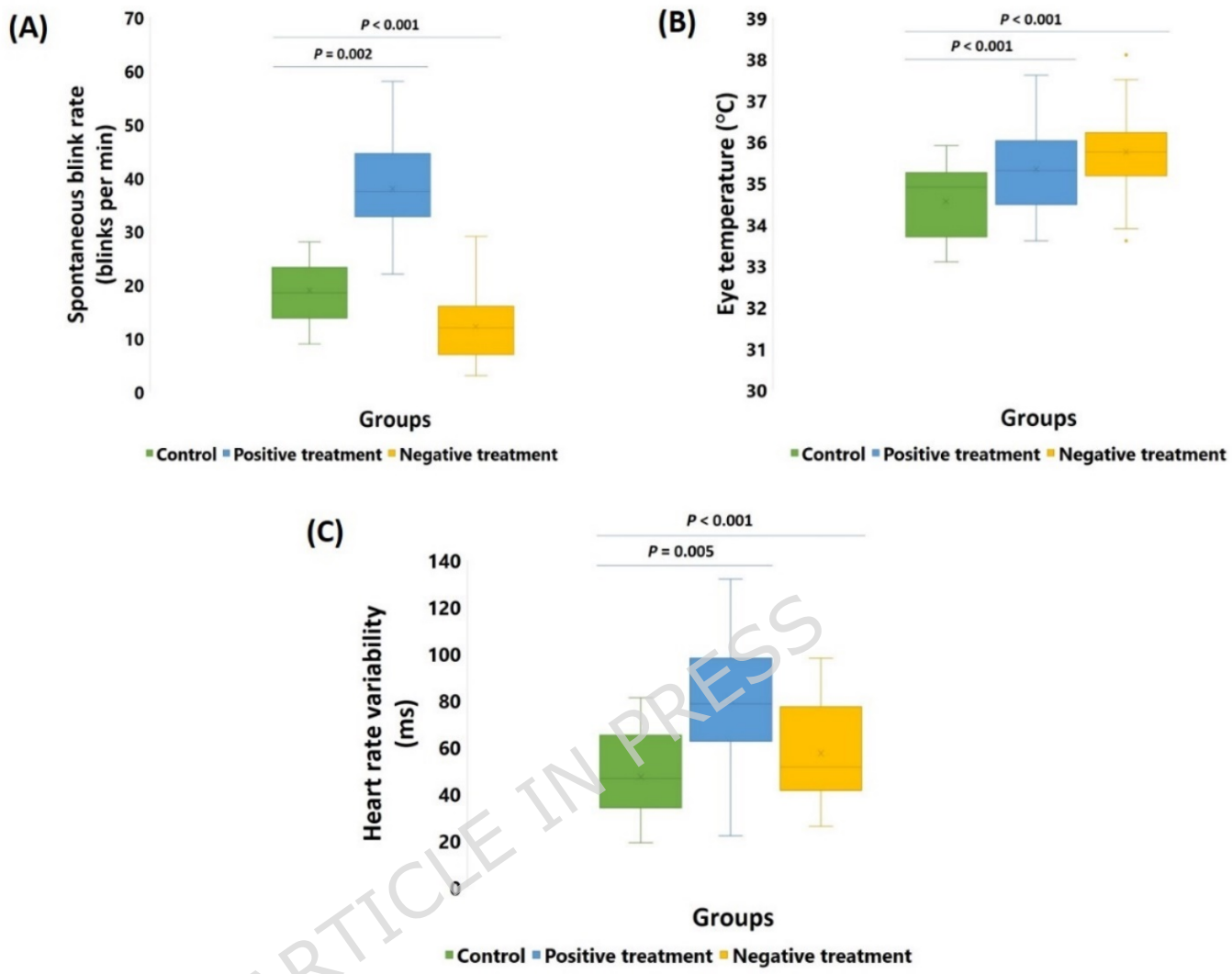
Variables	Control	Positive treatment	Negative treatment
Spontaneous blink rate (blinks per min)	18.94 ± 5.83	37.94 ± 8.37	12.20 ± 5.89
Change in spontaneous blink rate (blinks per min)	-	19.0±6.62	6.79±3.17
Eye temperature (°C)	34.55 ± 0.93	35.33 ± 1.02	35.75 ± 0.99
Change in eye temperature (°C)	-	0.81±0.56	1.22±0.57
RMSSD (ms)	47.26 ± 17.52	80.23 ± 27.23	57.35 ± 20.30
Change in RMSSD (ms)	-	37.73±15.95	10.73±7.04

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2.4 Comparison of spontaneous blink rate, eye temperature, and heart rate variability between control conditions and positive and negative treatments

Compared to control, the spontaneous blink rate increased significantly with the positive treatment (from 18.94±5.83 to 37.94±8.37 blinks per min, p=0.002), and decreased significantly with the negative treatment (from 18.94±5.83 to 12.20±5.89 blinks per min, p<0.001). Eye temperature significantly increased during both negative treatment (34.55±0.93 to 35.75±0.99°C, p<0.001) and positive treatment (34.55±0.93 to 35.33±1.02°C, p<0.001) compared to the control. Heart rate variability (RMSSD) was significantly higher with the positive treatment compared to the control (47.26±17.52 to 80.23±27.23 ms, p=0.005), Figure 2.

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Figure 2: (A) Spontaneous blink rates (blinks per minute), (B) eye temperatures (°C), and (C) heart rate variability (RMSSD) of donkeys during control, positive, and negative treatment are shown as box plots. Heart rate variability (HRV) was quantified using the root mean square of the successive differences between beat-to-beat intervals (RMSSD), a time-domain HRV metric reported in milliseconds. P-values were generated using linear mixed models.

2.5 Repeated-measure linear regression models

2.5.1 Relationship between change in spontaneous blink rate and heart rate variability (RMSSD)

A positive association was found between change in SBR and RMSSD (estimates = 1.32, t = 5.44, p < 0.001). Additionally, the positive treatment predicted a higher change in RMSSD (estimates = 10.82, t = 2.82, p = 0.006) relative to negative treatment (Table 3). A greater change in SBR and positive treatment was associated with a higher RMSSD level, Figure 3.

Table 3: Change in heart rate variability (ΔRMSSD) and spontaneous blink rate during positive and negative treatment. A table 3 displaying a repeated-measure linear regression model is presented. Negative treatment was used as the reference category for the treatment variable. Heart rate variability

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(HRV) was quantified using the root mean square of the successive differences between beat-to-beat intervals (RMSSD), a time-domain HRV metric reported in milliseconds.

Variables	Estimates	95% Confidence Interval	Standard error	t value	F value	p value
Intercept	1.72	-2.97 – 6.43	2.42	0.71	-	0.47
Change in spontaneous blink rate	1.32	0.85 – 1.79	0.24	5.44	29.60	<0.001
Treatment (Positive)	10.82	3.32 – 18.28	3.83	2.82	7.95	0.006

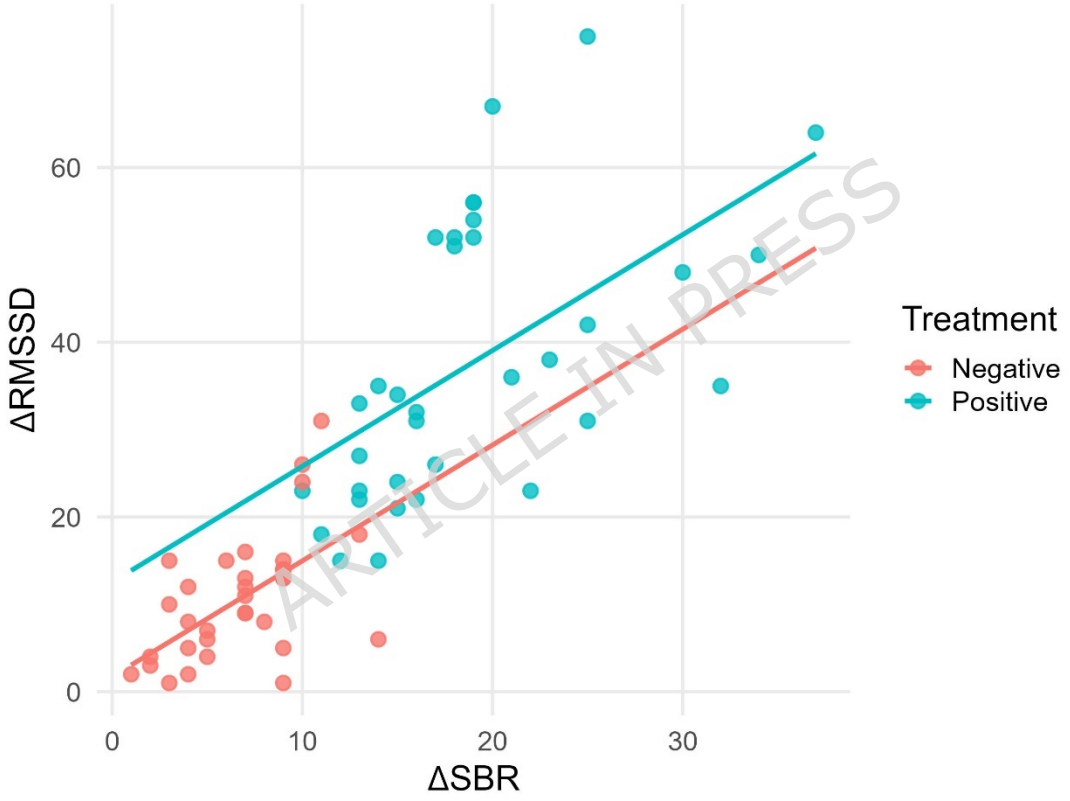


Figure 3: Change in heart rate variability (ΔRMSSD) and spontaneous blink rate (ΔSBR) during positive and negative treatments. Heart rate variability (HRV) was quantified using the root mean square of the successive differences between beat-to-beat intervals (RMSSD), a time-domain HRV metric reported in milliseconds. The figure is generated using the repeated-measure linear regression model.

2.5.2 Relationship between change in eye temperature and heart rate variability (RMSSD)

A negative association was found between changes in ET and RMSSD (estimates = -12.76, t = -5.93, p < 0.001). Additionally, the positive treatment predicted a higher change in RMSSD (estimates = 21.66, t =

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9.11, $p < 0.001$) relative to negative treatment (Table 4). A lower change in ET and positive treatment were associated with higher RMSSD level, Figure 4.

Table 4: Change in heart rate variability (ΔRMSSD) and eye temperature during positive and negative treatment. A table displaying a repeated-measure linear regression model is presented. Negative treatment was used as the reference category for the treatment variable. Heart rate variability (HRV) was quantified using the root mean square of the successive differences between beat-to-beat intervals (RMSSD), a time-domain HRV metric reported in milliseconds.

Variables	Estimates	95% Confidence Interval	Standard error	t value	F value	p value
Intercept	26.43	20.24 – 32.62	3.16	8.34	-	<0.001
Change in eye temperature	-12.76	-16.97 – - 8.54	2.15	-5.93	35.18	<0.001
Treatment (Positive)	21.66	17.02 – 26.37	2.37	9.11	82.99	<0.001

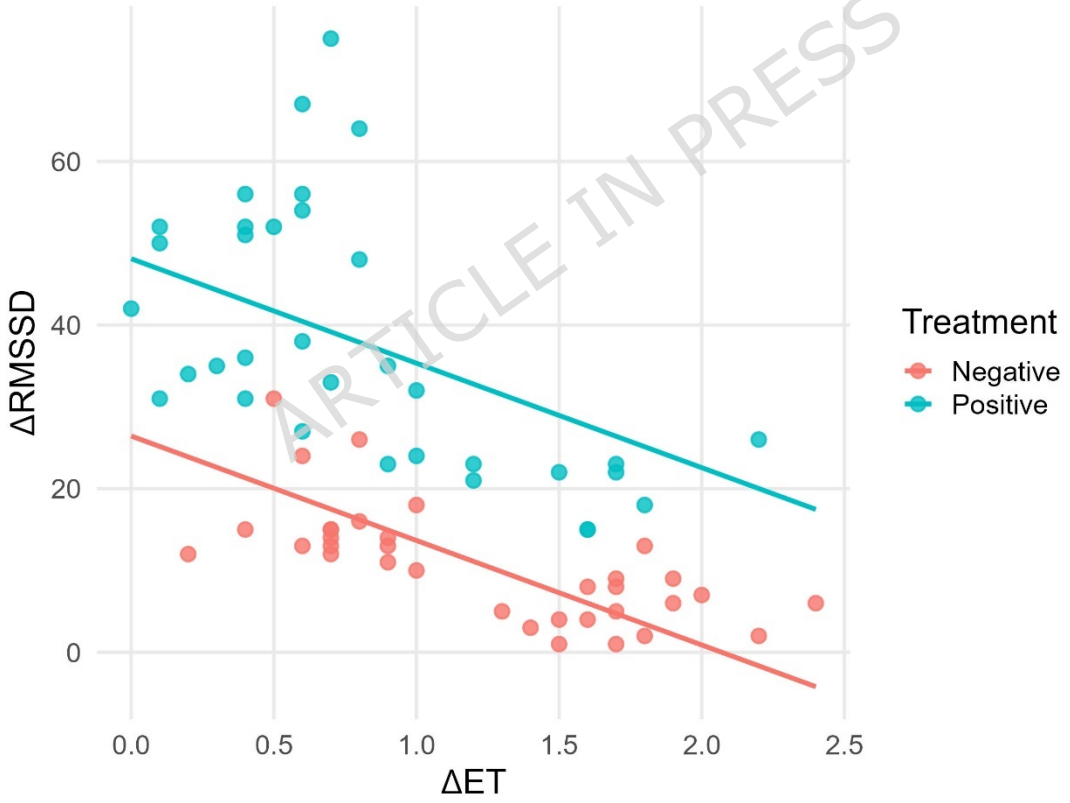


Figure 4: Change in heart rate variability (ΔRMSSD) and eye temperature (ΔET) during positive and negative treatment. Heart rate variability (HRV) was quantified using the root mean square of the successive differences between beat-to-beat intervals (RMSSD), a time-domain HRV metric reported in milliseconds. The figure is generated using the repeated-measure linear regression model.

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3 Discussion

Quantitative measures of emotion are essential for evaluating animal emotions and welfare. However, there is still a very long way to go to make accurate inferences about physiology and emotional state in animals [51]. In our study, we used SBR, ET, and HRV to measure donkeys' physiological responses to positive and negative stimuli. We tested donkeys with positive and negative treatments and measured their response in terms of SBR, ET, and HRV. Our results showed an increase in ET and a decrease in SBR in response to a negative treatment (clipper sounds), which is similar to previous studies in horses [10,13,40,41]. However, both ET and SBR increased in response to the positive treatment (provision of high-quality lucerne). When we calculated the change in each parameter (control minus positive and control minus negative treatment), the direction of change in each variable became more obvious. In response to positive treatment, the change in HRV and SBR was higher, but the change in ET was lower, while during negative treatment, the change in HRV and SBR was lower, and the change in ET was higher. This suggests that these measures capture different aspects of the emotional response. HRV and SBR are likely more related to autonomic regulation, attention, and emotional valence, whereas ET mainly reflects arousal or sympathetic activation. Thus, ET may rise during both positive and negative high-arousal events, with larger increases in negative or threatening situations. These findings suggest that SBR, ET, and HRV are promising measures of stress and positive emotions in donkeys, similar to humans [21], horses [10,26], dogs [9,34,45], goats [31,44], and cattle [35]. We propose that instead of considering absolute values for ET, SBR, and HRV, we should assess donkey emotions using the change in ET, SBR, and HRV.

Although donkeys may appear behaviorally stoic and show limited outward signs of their emotional state, this does not indicate reduced emotional or physiological sensitivity. Our findings show that donkeys exhibited measurable physiological responses to both positive and negative treatments in terms of SBR, ET, and HRV, which are similar to those of horses [10,13,28], dogs [9,34,45], humans [21], goats [31,44], sheep [32], cats [33], and cattle [35,46]. Donkeys are known for their stoicism and display a reduced flight response and increased freeze behaviors compared to horses [48,49]. It is often believed that donkeys are stoic and feel less pain or stress than horses. This perception is supported by their tendency to hide signs of pain, making it difficult to assess their well-being [50]. However, this is due to a lack of understanding of donkeys and their emotional states. Our study suggests that donkeys are not less sensitive than other animals, although they may show less overt behavioral change [52]. Moreover, we know that owner recognition of donkey emotions and feelings of pain benefits their welfare via improved management practices [20]. Furthermore, as donkeys may show few outward signs of stress, owners should not assume a quiet donkey is unaffected. Instead, they should consider whether factors such as loud noises, unfamiliar equipment, rough handling, isolation, or abrupt routine changes could be stressful even without obvious behavioral responses. Awareness of these subtle physiological changes can help owners detect stress sooner and adjust handling or management to improve welfare. Therefore, a better understanding of physiological indicators of donkey emotions and subsequent improved education of donkey owners may help improve their welfare in a challenging working environment.

We found decreased SBR in response to negative treatment (a stressful situation). This is similar to previous studies in horses [10,13,53] and humans [21]. A possible explanation for the increase in SBR during the positive treatment is that SBR is influenced by dopaminergic activity [54]. In monkeys (*Macaca fascicularis*), dopamine agonists increased SBR, whereas dopamine antagonists reduced SBR [25]. However, the present study did not measure dopamine, so this mechanism should be interpreted cautiously. Horses appear to respond in a similar way. Crib-biting is a common stereotypy in horses and is associated with stress, and crib-biting horses demonstrated lower SBR (15 to 11 blinks per min) than control animals. Additionally, a decrease in SBR was observed in horses during the initial one-minute treatment (clipping) compared to baseline (11 to 6 blinks per minute) [10]. This suggests that the reduced SBR may be the

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characteristic of a fight or flight response and increased sympathetic nervous system activity, allowing horses to visually fixate on the stimulus before responding accordingly [10]. In this study with horses [10], after an initial decrease in SBR in response to the negative stimulus, SBR started increasing. This was not observed in our study with donkeys. This may be because the horse researchers recorded SBR for an extended period of time (10 minutes), allowing the horses to become habituated to clipper sounds. Alternatively, there may be a difference between horses and donkeys in this regard. In comparison, during a startle test with horses involving tossing a ball, SBR decreased (26 to 13 blinks) compared to the control [13]. Our study demonstrates that SBR decreases in response to a stressful or startling situation. Furthermore, human research has shown a reduction in SBR during periods of attention and cognitive efforts [21]. Potentially threatening stimuli may require cognitive effort to verify them as a threat, thus reducing SBR. However, further investigation of changes in SBR over varying time spans is needed to identify patterns in response to stressful or startling stimuli in donkeys.

We found increased ET in response to negative treatments (a stress situation) similar to previous studies in horses [26–29]. Change in ET during positive experiences has yet to be investigated in horses, although it has been assessed in other species. Goats' eye temperature increased from 36.8 to 37.9 °C during a 1-minute stress situation caused by a person making noise and waving their arms in the animal pen. However, dogs showed an increase in ET (36.7 to 37.3 °C) during feeding (a high arousal, positive event) as compared to the control [9]. We know that high arousal situations, irrespective of whether they are positive or negative, can result in increased ocular vasodilation, leading to higher ET [9,30,55]. Interestingly, the change in ET during the negative treatment (1.22±0.57°C) was greater than positive treatment (0.81±0.56°C) in our study. Therefore, we propose that absolute ET is a measure of arousal, whereas change in ET is the measure of positive and negative emotions in donkeys. Since negative emotions are implicated in 'fight-or-flight' responses to threatening stimuli, we might expect quicker mobilization of the physiological responses underlying temperature changes to negative versus positive contexts [56]. Ultimately, the magnitude and time course of emotionally induced changes in ET may offer a means to examine emotional responses in relation to arousal and valence. Moreover, in agreement with our study, ET was higher (35.9 to 36.8 °C) after the novel object test (fear test) compared to basal ET in horses [28]. An increase in ET was also seen in response to a number of stressful conditions, for example, clipping (an increase of 3.5°C) [55], and with an over-tightened noseband (an increase of between 0.44 and 1.8 °C) in horses [26,57]. It is suggested that an increase in ET was caused by increased blood flow to the eyes as a sympathetic nervous system response [36]. Furthermore, an increase in ET was also positively correlated to salivary cortisol in response to stress in horses [29]. Therefore, ET should be interpreted as an indicator of physiological arousal or autonomic activation. In potentially negative contexts, changes in ET may help assess stress-system activation when interpreted alongside other physiological and behavioral measures.

We found an increase in RMSSD in response to positive treatment and a reduction in HRV in response to negative treatment (stressful situation), which is consistent with previous studies in horses [40–43]. HRV is a sensitive measure of both physical and emotional stress in horses [58], and it is a commonly used autonomic nervous system (ANS) correlate, estimating the sympathovagal balance as a psychophysiological marker of emotion regulation across species. It is a sensitive measure of both physical and emotional stress in horses [58], and it is a commonly used autonomic nervous system (ANS) correlate, estimating the sympathovagal balance as a psychophysiological marker of emotion regulation across species [40]. RMSSD was higher in response to positive calls (65ms) compared to negative ones (45ms) in goats [44]. Similarly, RMSSD can also vary seasonally as Przewalski's horses (*Equus ferus przewalskii*) exhibited low RMSSD (70 ms) during winter, when resources are limited, indicating decreased metabolic rates coupled with stress, compared with summer (RMSSD = 250 ms) [41]. Furthermore, horses showed a decrease in RMSSD (46.3 to 15.8ms) during novel object and handling tests [42], indicating a shift of the autonomic balance towards a sympathetic dominance. The RMSSD decreased (76 to 46 ms) in response

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to training of young sport horses, and the lowest values were reached during mounting of a rider (23 ms) [43]. The most pronounced stress reaction occurred in response to the horses being mounted by a rider, a situation resembling an imminent threat under natural conditions. Furthermore, in horses, RMSSD remains high in interaction with familiar handlers as compared to unfamiliar handlers [40], similar to the change in RMSSD, which was high in donkeys when lucerne was provided for eating. Therefore, RMSSD is a valuable measure to quantify sympathovagal balance, which allows a more precise assessment of stress and positive emotions [40].

To our knowledge, this is the first report examining how SBR, ET, and HRV vary in donkeys exposed to stimuli intended to induce positive and negative emotional states. SBR is a relatively easy measure to examine, although the observer must be fairly close to the animal in question [10,13]. Monitoring changes in SBR can alert the observer to changes in the donkey's emotional state. A decrease in SBR may indicate a stressful situation for the donkey. This is particularly significant in low-resource settings where access to advanced technology may be limited. ET and HRV are important measures of the emotional state of a donkey, but they need very sophisticated and expensive equipment for monitoring. Moreover, ET should be interpreted carefully as it increases in both positive and negative treatment [9,55]. It is important to note that ET, SBR, and RMSSD may serve as measures of donkey emotions, but require context. SBR may be relevant to affective processes; its interpretation across positive, negative, and neutral contexts remains complex and requires further investigation. RMSSD was the only HRV metric analyzed in this study. Future studies should include a broader HR/HRV analysis, including heart rate, additional time-domain measures, frequency-domain indices, and non-linear measures, while carefully accounting for artefacts, respiratory influences, and ectopic beats. It would also be helpful if further research could include longer observations of donkeys during positive and negative treatment and after removal of the treatment, as the interplay between SBR, ET, and HRV in this period may reveal more about the importance of these measures for the assessment of stress and positive affective state. Future research should include longer observation periods during both positive and negative treatments, along with a washout period. Examining the interplay between SBR, ET, and RMSSD in this phase may clarify how useful these measures are for assessing stress and positive affective state. In relation to this, future studies should examine donkey behaviors, such as reluctance to move forward, ear position, and muzzle and nasal tension, as this study only included one behavior (SBR) and lacked a comprehensive behavioral investigation during treatments. As interest grows, it is important to develop a detailed understanding of their potential and limitations.

4 Conclusion

Our study showed that ET, SBR, and RMSSD varied in working donkeys exposed to positive and negative treatment. The positive treatment was associated with numerically greater changes in RMSSD and SBR, and a smaller change in ET. In contrast, the negative treatment was associated with numerically smaller changes in RMSSD and SBR, and a larger change in ET. Importantly, SBR may provide a simple and practical behavioral indicator of donkey responses to positive and negative treatment, without the requirement for sophisticated equipment. This is particularly significant in low-resource settings where access to advanced technology may be limited. Furthermore, our study challenges the notion that donkeys are stoic and experience less pain or stress than horses. We found that donkeys respond to positive and negative treatments in a similar way to other species such as horses, goats, dogs, and cattle.

5 Methods

5.1 Ethical Approval

This research project was approved by the Animal Research Ethics Sub-Committee of City University of Hong Kong (No. P9610463).

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5.2 Sample size estimation

The sample size was estimated using G*Power 3.1 software [59]. Previous research reported that SBR decreases significantly (11.1 to 6.2 blinks per minute) under minor stress (negative) situations in horses [10]. Conversely, ET increases significantly (32.3 to 35.9 °C) under acute stress (negative) situations as compared to control situations in horses [29]. These studies were used to support the biological plausibility of detecting treatment-related changes in donkeys, but not to derive the effect size for the power analysis. We hypothesized that changes to the SBR, ET, and HRV of donkeys exposed to positive and negative stimuli would be comparable to previous horse studies [10,29]. Therefore, the input parameters set for G*Power were alpha ($\alpha = 0.05$), power (0.95), and effect size ($\rho = 0.5$), leading to a sample size estimation of 34 donkeys for the current research.

5.3 Animals

Thirty-four Pakistani working donkeys (age 6.05±0.31 years; Jennies 20, Jacks 14) were used for the study. All the donkeys were working animals and were therefore accustomed to human presence. They all remained in their home stable during the experiment and were housed individually. Donkeys were physically fit with a good body condition score (2.88±0.40 out of 5) according to The Donkey Sanctuary guidelines [60]. Donkeys were assessed by an equine veterinary surgeon at rest for overall health status based on The Brooke's SEBWAT welfare assessment tool [15]. Healthy donkeys were recruited based on guidelines recommended from previous research [15]. Verbal consent was obtained from the donkey owners prior to the commencement of data collection. Owners reported that none of the donkeys were given any medication during the testing period or for one month before data collection.

5.4 Study design and data collection

A randomized crossover trial was used to assess physiological responses to three treatments, with a one-hour washout period. Prior to undertaking the experiments, the researcher (SSUHB) habituated each donkey to his presence. This was achieved by spending two hours daily with the donkeys for one and a half months to minimize any effect of an unfamiliar person on the animals' physiological responses. The researcher also visited the donkeys with the relevant research equipment to habituate them to both the equipment and the researcher. We chose the SBR as the habituation criterion. When we noticed that the blink rate (blinks per min) stayed constant in the presence and absence of the experimenter and research equipment, we concluded that habituation had been achieved. Following habituation, donkeys were subject to three treatments: positive, negative, and control. The positive treatment was the provision of high-quality green lucerne (food, 8-liter bucket). Donkeys in this area like fresh lucerne, but they usually don't have access to it because it's expensive. The negative treatment was the sound of hair clippers held close to the donkey's right ear. Positive and negative treatments were chosen based on previous research in dogs [9] and horses [10]. Donkeys standing calmly in their usual living environment without positive or negative treatment and any other interaction were used as the control. On data collection day, donkeys were not working to prevent confounding effects from working conditions. Each donkey was used as its own control to evaluate the relationship between variables within each treatment period. Data on SBR, ET, and HRV were collected for each donkey during each treatment. The treatment sequence was randomly assigned by generating random numbers between one to three using Microsoft Excel. Number one was designated as the control treatment, number two as the positive treatment, and number three as the negative treatment. As the trials followed a randomized crossover design, each donkey received the treatments (control, positive, and negative) in a random order on the same day (single session). All measurements were taken between 10:00 and 14:00 each day to avoid any confounding effects of diurnal variation in physiological recordings. Data were collected during each treatment for three minutes [13]. This

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resulted in nine minutes of SBR, ET, and HRV data for post-hoc analysis. Figure 5 shows the experimental setup.

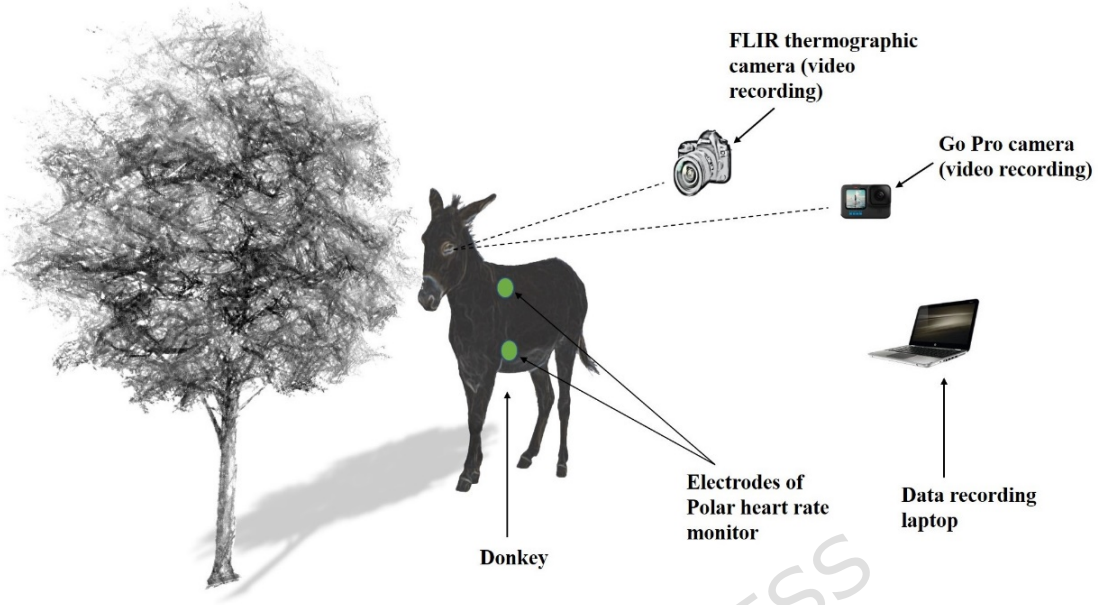


Figure 5: Thirty-four donkeys were assessed for physiological responses to control, positive, and negative treatments. Donkeys were filmed with a FLIR thermographic camera (FLIR T530, Teledyne FLIR Systems Inc., Wilsonville, USA) to assess eye temperature, a GoPro camera to record spontaneous blink rate, and wore a Polar heart rate monitor to measure heart rate variability (RMSSD - root mean square of the successive differences between beat-to-beat intervals).

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5.5 Spontaneous blink rate

The donkey's SBR was recorded by filming its eyes using a GoPro HERO Black camera at 30 frames per second. The videos were transferred to a laptop and analyzed for SBR, similar to a previous research [10]. Measurements of SBR were made using coding of the offline videos by two coders (coder-1 and coder-2). Each coder counted the number of blinks per minute. SBR was reported as blinks per minute for each donkey under each test condition.

5.6 Eye temperature

A standardized infrared thermography protocol for ET (°C) was used [61]. Each donkey was filmed under the shade using a thermographic camera (FLIR T530, Teledyne FLIR Systems Inc., Wilsonville, USA), placed at 90° to the sagittal plane of the eye and at a distance of one meter [61]. The left eye was selected following the experimental protocol adopted for horses [10]. The maximum temperature at the medial canthus of the eye was recorded using FLIR Tools+ software, as it gives the most reliable temperature measurement [32]. The camera's distance from the donkey was measured using laser technology fitted within the FLIR T530 camera. Thermographic videos were transferred to a laptop via SD cards. The data were extracted from videos using FLIR-Tools+ software. The maximum temperature at the medial canthus of the eye was recorded as it gives the most reliable temperature measurement [32]. The ambient temperature and humidity were measured using a portable weather station (LondiSun LS-204), and wind speed was measured using a Kestrel-1000 digital anemometer, as they may affect thermography in field conditions.

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5.7 Heart rate variability

Each donkey was fitted with a heart rate monitor (Polar Equine V800) to record beat-to-beat interval (R-R interval). HRV describes the variation in the time intervals between successive heartbeats. In this study, HRV was quantified using the root mean square of the successive differences between beat-to-beat intervals (RMSSD), a time-domain HRV metric reported in milliseconds [10]. The data collected from the Polar Equine V800 were exported to the laptop using the Polar FlowSync software. The data were analyzed using Kubios software (Kubios HRV Standard version 3.5.0, Kubios Oy, Kuopio, Finland), with a low threshold (0.35 s) for artifact correction according to previous research [10,23].

5.8 Statistical analysis

All the variables are presented as mean and standard deviation (mean ± SD). The Shapiro-Wilk test was used to test the normality of the data. SBR data extracted by coder-1 were used for the main statistical analysis because coder-1 was the primary trained observer who coded all videos. Coder-2 independently coded the videos to assess inter-observer agreement. The use of coder-1 data was considered appropriate because Bland–Altman analysis showed high agreement between coders, with minimal bias and narrow limits of agreement across the control, positive, and negative treatments. For SBR, Band-Altman plots were used to determine the level of agreement between coders (C1 and C2) and bias. Data were normally distributed according to the Shapiro–Wilk test ($p > 0.05$); therefore, raw, untransformed data were used for all analyses. Linear mixed-effects models (with an individual donkey as a random effect) were fitted to individually compare SBR, ET, and HRV (quantified as RMSSD) under the positive or negative treatment with the control. The order of the treatment application was included in the model as a fixed effect to account for the randomized crossover nature of the study. The change in each parameter (SBR, ET, and HRV) was calculated (control minus positive and control minus negative treatment). The changes in SBR and ET were compared with an established physiological indicator of autonomic nervous system activity associated with stress and emotional arousal (change in HRV [10]) in animals using two repeated-measure linear regression models. An individual donkey was included as a random effect. In the first model, change in HRV, quantified as RMSSD, was the outcome variable, with change in SBR and treatment included as fixed effects. In the second model, change in RMSSD was the outcome variable, with change in ET and treatment included as fixed effects. All statistical analyses were conducted using R packages tidyverse, lme4, lmerTest, and RStudio with R version 4.2.3 [62].

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Declarations

Ethical Approval

This research project was approved by the Animal Research Ethics Sub-Committee of City University of Hong Kong (No. P9610463). All experimental procedures involving animals were carried out in accordance with relevant institutional and national guidelines and regulations for the care and use of animals. This study is reported in accordance with the ARRIVE guidelines (https://arriveguidelines.org).

Data availability statement

The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

Author contributions statement

SSUHB: Conceptualization, Investigation, Methodology, Data curation, Formal data analysis, Writing–original draft, Writing – review & editing. AGM: Supervision, Conceptualization, Investigation, Methodology, Writing – review & editing. PVS: Supervision, Writing – review & editing. RSVP: Supervision, Conceptualization, Investigation, Methodology, Writing – review & editing. All authors were involved in the research, contributed to the preparation, and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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Figure legends

Figure 1: Bland and Altman plots showing the agreement and bias of the spontaneous blink rate (blinks per min) of donkeys between coders (C1 and C2) during the (A) control (B) positive, and (C) negative treatment. The bias (green continuous line) and limits of agreement (orange dotted lines) are indicated on each plot.

Figure 2: (A) Spontaneous blink rates (blinks per minute), (B) eye temperatures (°C), and (C) heart rate variability (RMSSD) of donkeys during control, positive, and negative treatment are shown as box plots. Heart rate variability (HRV) was quantified using the root mean square of the successive differences between beat-to-beat intervals (RMSSD), a time-domain HRV metric reported in milliseconds. P-values were generated using linear mixed models.

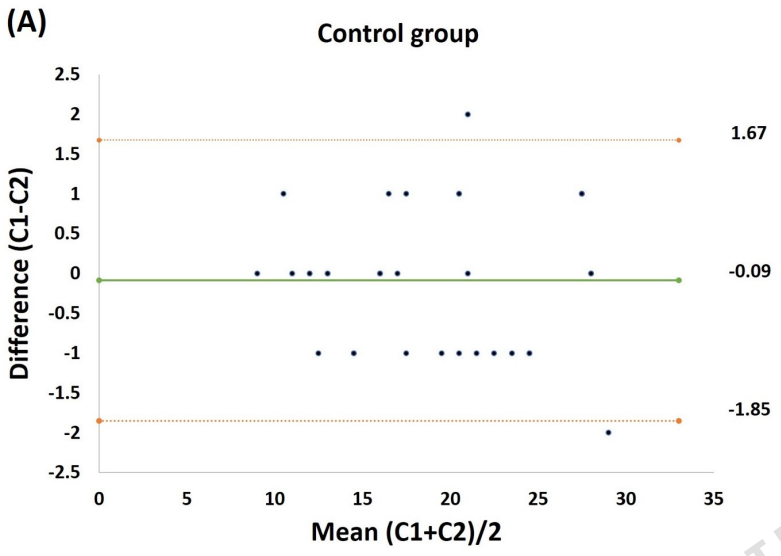
Figure 3: Change in heart rate variability (ΔRMSSD) and spontaneous blink rate (ΔSBR) during positive and negative treatments. Heart rate variability (HRV) was quantified using the root mean square of the successive differences between beat-to-beat intervals (RMSSD), a time-domain HRV metric reported in milliseconds. The figure is generated using the repeated-measure linear regression model.

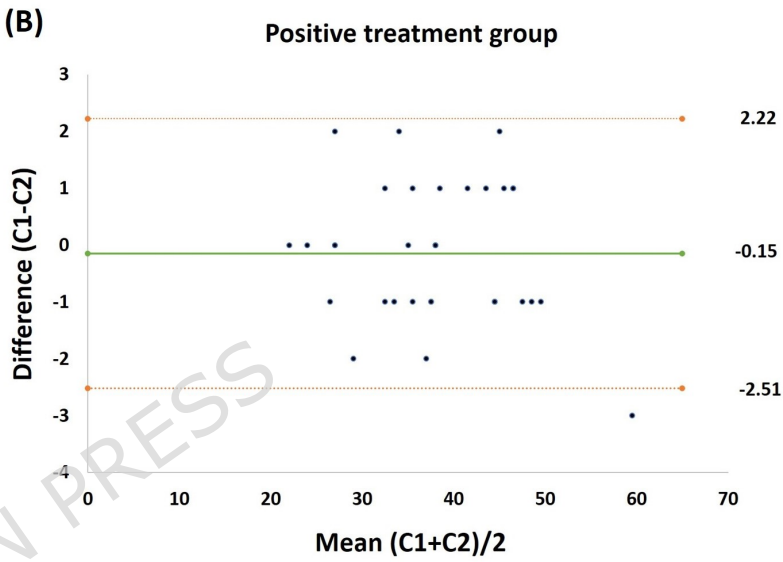
Figure 4: Change in heart rate variability (ΔRMSSD) and eye temperature (ΔET) during positive and negative treatment. Heart rate variability (HRV) was quantified using the root mean square of the successive differences between beat-to-beat intervals (RMSSD), a time-domain HRV metric reported in milliseconds. The figure is generated using the repeated-measure linear regression model.

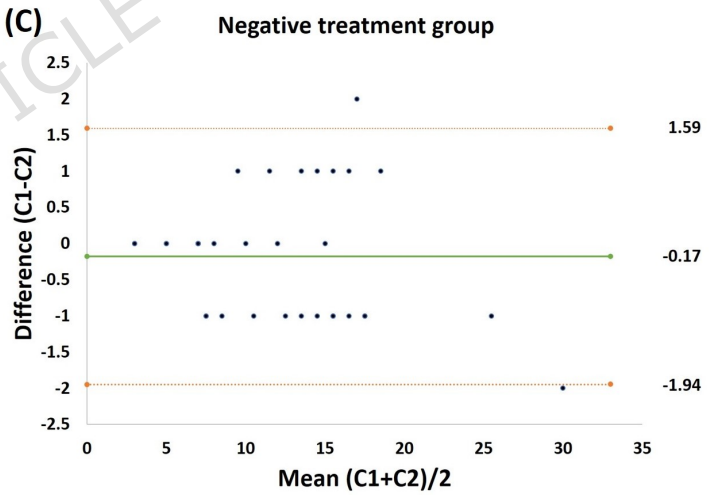
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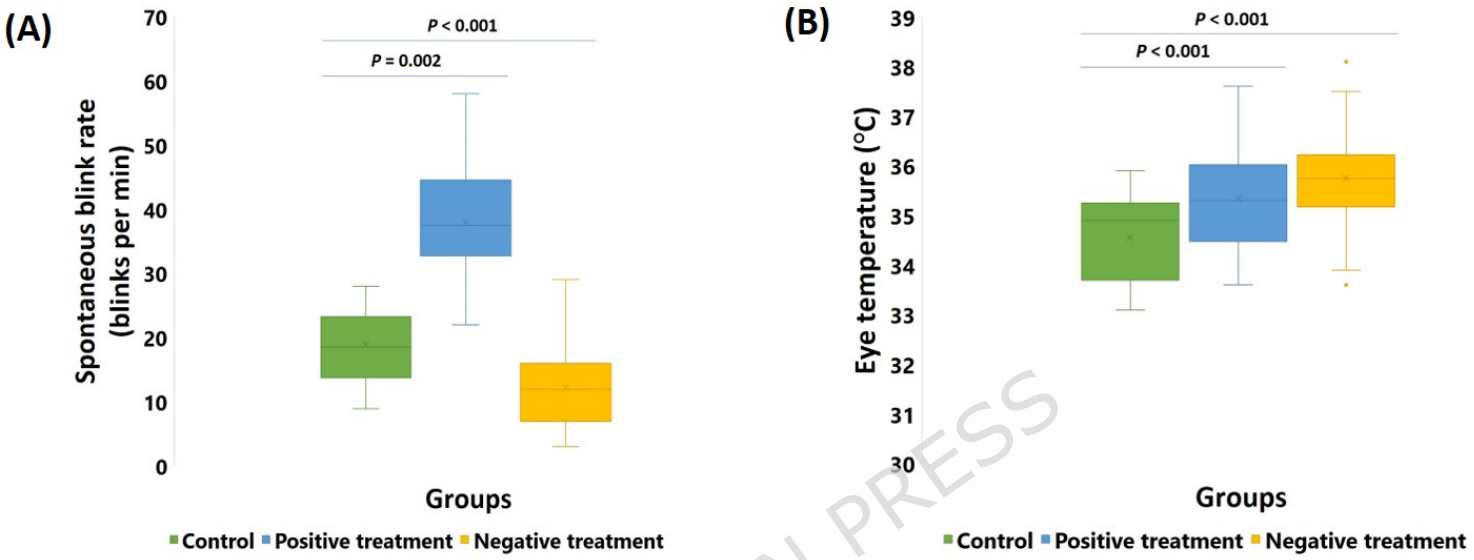


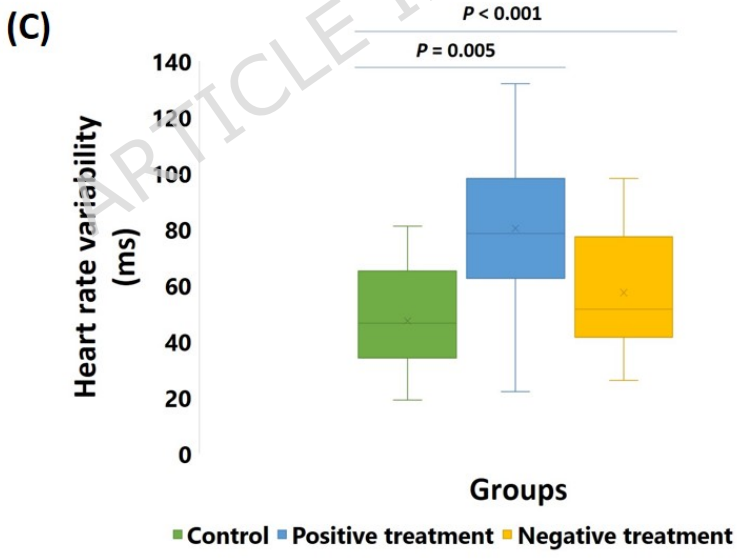




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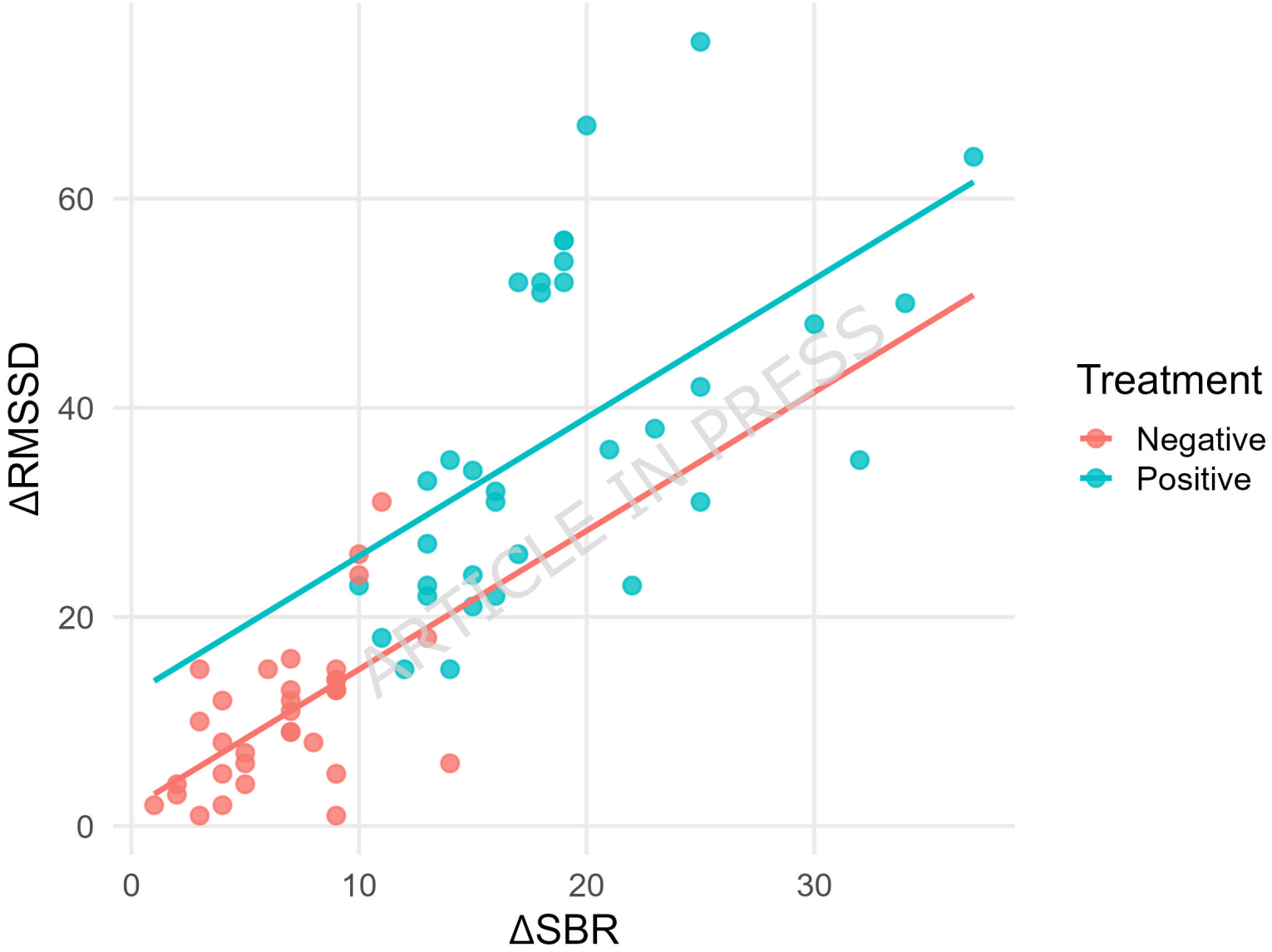
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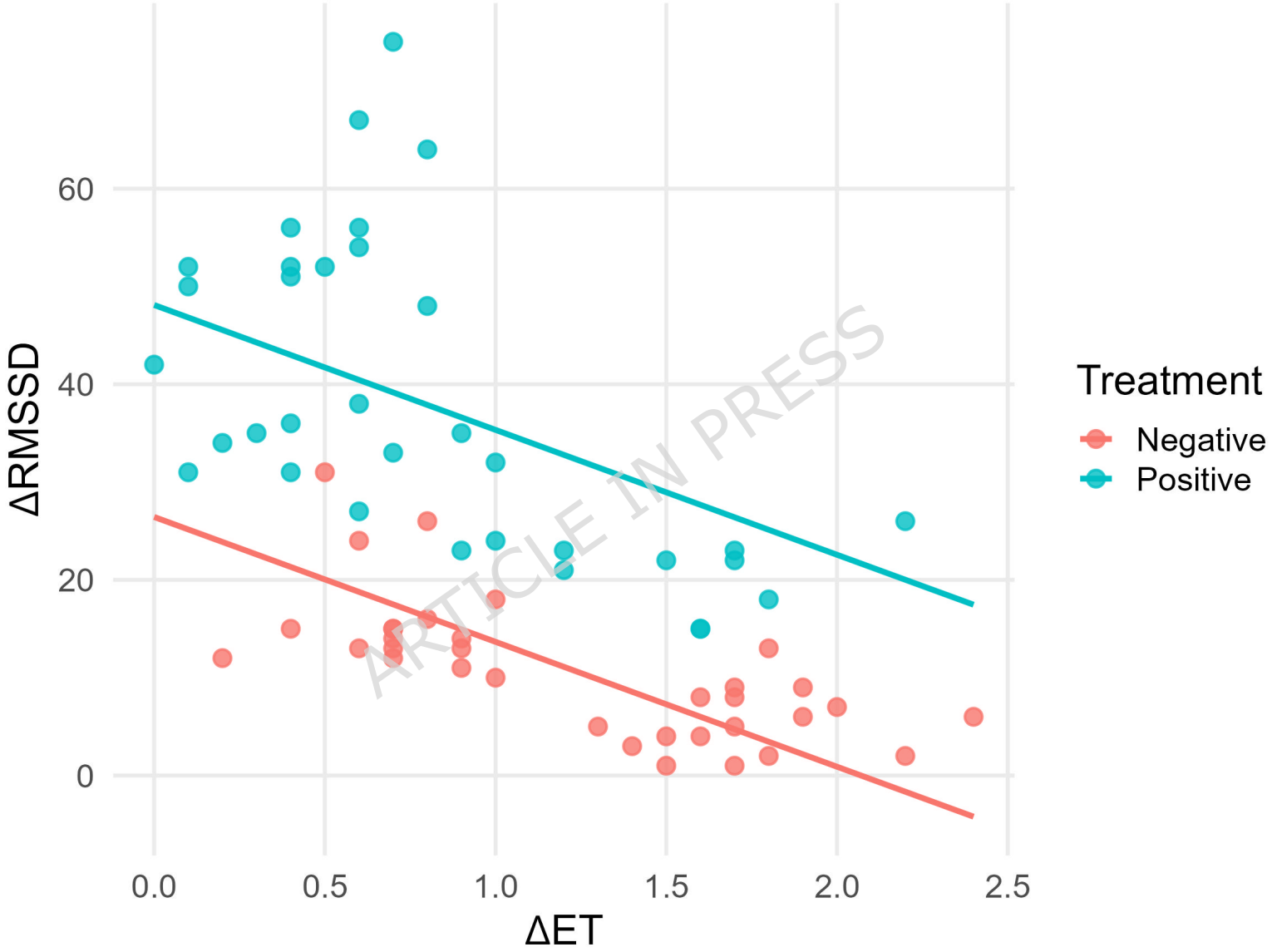
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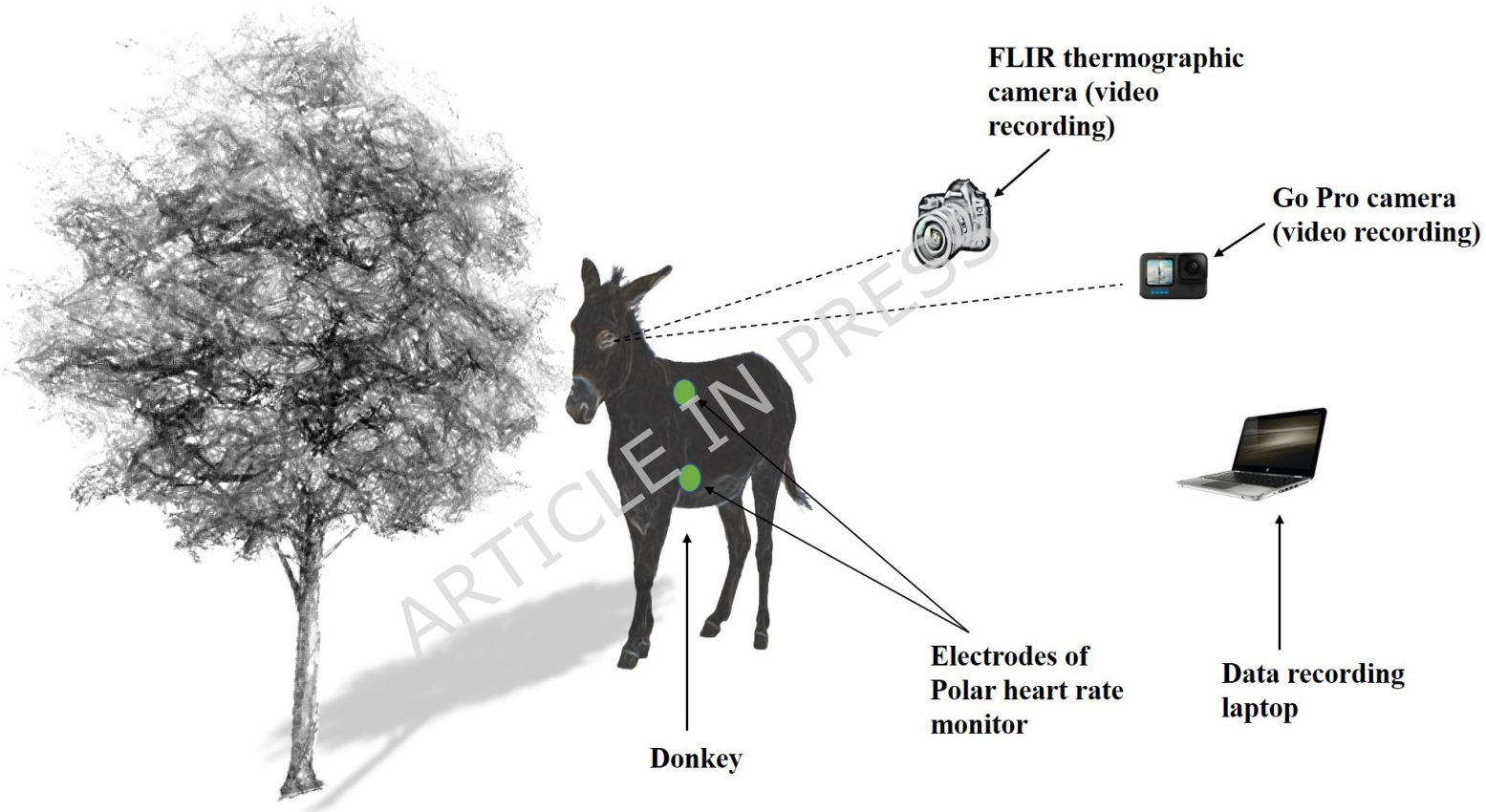


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